

Day 4 - Dataset Design & Quality Expectations

Expected Dataset Structure

Column	Description
Received Signal	Analog signal value received at the receiver
NTU	Turbidity level of the water
Transmitted Bit	Actual bit label (0 or 1)
Metadata (Optional)	Sample ID, timestamp, place,

Possible Dataset Risks :

- Missing Data → Missing signal or NTU values.
- Wrong Labels → Incorrect bit labels (0/1).
- Class Imbalance → More samples of one bit than the other.

Dataset Risk Checklist :

- Check missing values
- Check duplicate records
- Verify bit labels (0/1)
- Check class balance
- Check NTU coverage → Some turbidity levels may have very few samples.
- Check data leakage → Similar samples appearing in both train and test sets.
- Validate train-test split :

Train Set (70%) → Model learning.
Validation Set (15%) → Model tuning.
Test Set (15%) → Final evaluation.

